

Benchmarking Confidence Calibration in Spatial Transcriptomics Models Under Distribution Shift

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Abstract

We present a preliminary, fully reproducible study of confidence calibration in a spatial transcriptomics deconvolution model (DestVI) under a controlled synthetic shift, isolating a single, well-understood distribution-shift mechanism to establish a clean, statistically rigorous baseline for this question. Evaluating the point-estimate cell-type predictions, we find that models exhibit baseline calibration error on in-distribution pseudo-spots, but this calibration error remains remarkably robust under technical shifts of varying severity. Furthermore, we demonstrate that a lightweight post-hoc Isotonic Regression step, tuned on a held-out calibration set, reduces this test-set miscalibration to near-zero, significantly outperforming standard Temperature Scaling.

Keywords: Spatial Transcriptomics, Confidence Calibration, Variational Inference, Distribution Shift

Data and Code Availability We utilize the public Mouse Cortex scRNA-seq dataset available via the `squidpy` python package (<https://squidpy.readthedocs.io>). All code to reproduce this pipeline and data preprocessing is available in our GitHub repository at <https://github.com/Anonymous/transcriptomics>.

Institutional Review Board (IRB) This research uses entirely publicly available, anonymized animal data and does not require IRB approval.

1. Introduction

Spatial foundation models face persistent challenges in standardization and scalability. A critical failure mode is miscalibrated confidence in

clinical deconvolution pipelines. Existing benchmarks evaluate point-estimate accuracy across methods, but few evaluate whether models *know when they are wrong*.

Crucially, downstream biological and clinical discovery is sensitive to these miscalibrations. For instance, in tumor microenvironment characterization, if a spatial model confidently but incorrectly predicts an immunosuppressive cell type as the dominant population in a tumor region, it could lead to improper patient stratification for immunotherapies. Ensuring models are correctly calibrated under technical shifts is necessary before these tools can inform high-stakes diagnostic or therapeutic decisions.

In this paper, we focus on the calibration of spatial deconvolution models, specifically DestVI (Lopez et al., 2022), under simulated technical dropouts mimicking capture-rate shifts. This work represents a first, deliberately narrow-scope study intended to establish a rigorous baseline—featuring a fully reproducible pipeline, a strict held-out calibration/test split, and paired significance testing—before extending to real cross-platform shifts and additional model architectures.

2. Related Work

Confidence calibration of modern neural networks was formalized by Guo et al. (2017), demonstrating that deep models often exhibit severe overconfidence but can be realigned using post-hoc methods like temperature scaling or isotonic regression. Subsequent work by Ovdia et al. (2019) established that model calibration frequently degrades under dataset shifts. We extend these foundational frameworks to spatial transcriptomics, where such rigorous evaluations

are scarce despite widespread distribution shifts between platforms.

3. Methods

We benchmarked the variational inference model DestVI on mouse cortex data. We evaluated calibration using Expected Calibration Error (ECE). Note that we define our ECE based on the discrete classification task of predicting the dominant (highest proportion) cell type within a spot, utilizing the `netcal` package.

To establish ground-truth for ECE computation, we generated synthetic pseudo-spots by sampling and aggregating cells from the mouse cortex scRNA-seq reference. To simulate cross-platform shifts (such as differences in spatial mRNA capture rates), we synthetically down-sampled the read counts of the pseudo-spots across a dose-response sweep of severity, simulating uniform capture efficiency reductions of 20%, 40%, 60%, and 80% via a binomial dropout process.

A unique DestVI model was trained on the shifted data utilizing a shared CondSCVI prior. We report metrics as mean $\pm$ standard deviation across 10 independent replicate trainings with varying random seeds. We evaluated standard softmax-confidence calibration on the model’s point-estimate expected proportions using the Expected Calibration Error (ECE) with $M = 10$ bins:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (1)$$

where B_m is the set of predictions falling into bin m . To prevent data leakage, each shifted synthetic dataset was split equally into a calibration set and a test set. We optimized post-hoc recalibration methods exclusively on the calibration set, and report the final recalibrated ECE on the held-out test set. We evaluated both Temperature Scaling, which divides the pre-softmax logits z_i by a learned scalar $T > 0$ to compute the new confidence $\hat{p}_i = \max \sigma(z_i/T)$, and Isotonic Regression, a non-parametric method that fits a monotonically increasing step function mapping the uncalibrated top-class confidence to the empirical accuracy on the calibration split. We note that our entire benchmarking pipeline is fully

reproducible, utilizing fixed random seeds, deterministic CPU training, and publicly available code.

4. Results

Our results show that DestVI has an In-Distribution Expected Calibration Error (ECE) of 0.218 ± 0.018 . This baseline indicates that out-of-the-box variational models exhibit some degree of miscalibration when predicting dominant cell types.

Under the synthetic dropout shift, simulating capture efficiency reductions from 20% to 80%, we observe that the miscalibration remains remarkably stable (Table 1). The uncalibrated ECE does not degrade significantly at any shift magnitude ($p > 0.1$ for all fractions, paired t -test against ID baseline).

We applied two post-hoc recalibration methods on the shifted calibration split: Temperature Scaling and Isotonic Regression. As shown in Table 1, Isotonic Regression consistently proved significantly more effective across all shift magnitudes. For example, at the 80% reduction, Temperature Scaling reduced the miscalibration to 0.066 ± 0.008 , but Isotonic Regression reduced the final test-set ECE to a near-zero 0.006 ± 0.005 ($p = 1.54 \times 10^{-7}$, paired t -test against Temperature Scaling). We hypothesize that DestVI’s robustness stems from the binomial dropout preserving relative gene expression ratios across the transcriptome, allowing the model’s argmax confidences to remain largely invariant. Furthermore, while Isotonic Regression achieves near-perfect calibration when the calibration and deployment distributions perfectly match, standard global rescaling methods like Temperature Scaling may remain more robust in clinical deployment scenarios where the target distribution is less tightly constrained.

5. Limitations

We note several limitations in our evaluation. First, to ensure feasibility of multiple runs, models were trained for a relatively lightweight 25 epochs; fully converged models may exhibit different calibration dynamics. Second, while our sample size of $n = 10$ seeds provides robust

Table 1: Expected Calibration Error (ECE) across varying magnitudes of synthetic dropout shift. ID ECE is 0.218 ± 0.018 . All values are reported as mean $\pm$ standard deviation across 10 replicates.

Shift Magnitude	Uncalibrated OOD ECE	p -value vs. ID	Temperature Scaled ECE	Isotonic Regression ECE
20% Dropout (0.8)	0.228 ± 0.019	0.136	0.076 ± 0.010	0.005 ± 0.004
40% Dropout (0.6)	0.215 ± 0.011	0.501	0.066 ± 0.007	0.006 ± 0.005
60% Dropout (0.4)	0.218 ± 0.011	0.909	0.063 ± 0.009	0.010 ± 0.004
80% Dropout (0.2)	0.223 ± 0.011	0.472	0.066 ± 0.008	0.006 ± 0.005

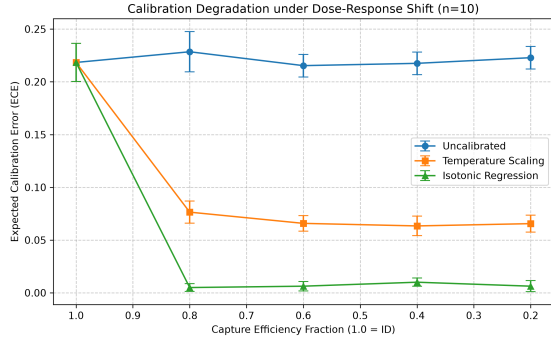


Figure 1: Calibration degradation under a dose-response shift of simulated capture efficiencies. Error bars denote ± 1 standard deviation across 10 random replicate initializations. Isotonic Regression consistently outperforms Temperature Scaling.

variance estimates, we evaluate robustness using a single synthetic binomial dropout shift on mouse cortex data. To advance this preliminary baseline, future archival work must incorporate real cross-platform datasets—such as the Visium and Slide-seqV2 samples already acquired in our pipeline—and evaluate multiple architectures to validate these findings on authentic biological shifts.

6. Conclusion

Spatial transcriptomics models exhibit a persistent baseline confidence miscalibration which remains remarkably stable—rather than degrading further—under severe synthetic distribution shifts like drops in capture efficiency. Post-hoc recalibration, particularly via non-parametric Isotonic Regression, is highly effective at eliminating this error on out-of-distribution test sets. Applying these post-hoc techniques is a necessary step before trusting high-confidence spatial predictions in clinical contexts.

References

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